INDRA GANESAN COLLEGE OF ENGINEERING

(An Autonomous Institution, Affiliated to Anna University, Chennai)

(Approved by AICTE, New Delhi & Accredited by NAAC)

Manikandam, Tiruchirappalli – 620 012

Tamil Nadu

REGULATIONS FOR UG DEGREE PROGRAMME
(For B.E. / B.Tech. Programme)

REGULATIONS – 2024

CHOICE BASED CREDIT SYSTEM (CBCS)

(For the Students admitted during the
Academic Year 2024-25 onwards)

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(Choice based Credit System)

(Common to all B.E. / B.Tech. Degree Programmes)

Regulations 2024 have been prepared in accordance with the guidelines given by the University Grants Commission, All India Council for Technical Education and affiliating Anna University incorporating the features of the Choice Based Credit System (CBCS). The Regulation 2024 is applicable to the candidates admitted to the first year Bachelor of Engineering (B.E.) / Bachelor of Technology (B.Tech.) Degree Programmes of the Institution from the academic year 2024-2025 onwards and academic year 2025-2026 for second year Lateral Entry students.

The regulations hereunder are subjected to amendments as may be decided by the Academic Council of the Institution from time to time. Any or all such amendments will be effective from such date and to such batches of students (including those already in the middle of the programme) as may be decided by the Academic Council.

1.DEFINITIONS

In these Regulations, unless the context otherwise specified:

1.1. **"Academic Autonomy"** means autonomy granted by the University Grants Commission (UGC) and Anna University, Chennai to Indra Ganesan College of Engineering, Tiruchirappalli in all aspects of conducting its Academic Programmes for promoting excellence.

1.2. **"Autonomous College"** means a College notified as an Autonomous college by the University, as per the Anna University Autonomous College Statute.

1.3. **"Commission"** means University Grants Commission.

1.4. **"Council"** means All India Council for Technical Education.

1.5. **"Statute"** means Anna University Autonomous College Statute.

1.6. **"University"** means Anna University, Chennai.

1.7. **"College"** means Indra Ganesan College of Engineering, Trichy.

1.8. **"Programme"** means under graduate degree programme. i.e., B.E. / B.Tech. Degree Programme.

1.9. **"Discipline / Branch"** means specialization in a programme like B.E. Degree Programme in Mechanical Engineering, B.Tech. Degree Programme in Agricultural Engineering, etc.

1.10. **"Course"** means a subject either theory or practical identified by its course title and code, and which is normally studied in a semester, for example, 'Basics of Electrical and Electronics Engineering'.

1.11. **"Choice-Based Credit System (CBCS)**" is a flexible system of learning that permits students to

- Learn at their convenient pace.
- Choose electives from a wide range of elective courses offered by the departments.
- Adopt an inter / trans-disciplinary approach in learning and
- Make the best use of the expertise of available faculty.

1.12. **"Semester"** means a term of study consisting of a minimum of 90 working days (including examinations), generally with eight 'CONTACT HOURS' per day. The Semester scheduled between July and December shall be called 'Odd Semester' (I, III, V, VII), and that scheduled between January and May, shall be called 'Even semester' (II, IV, VI, VIII).

1.13. **"Contact Hours"** means duration of Lecture, Tutorial, Practical or Project Work, as applicable.

1.14. **"L – T – P – C"** means Lecture, Tutorial, Practical and Credits respectively.

1.15. **"Credit"** means a numerical value allocated to each course to describe the student's contact hours required per week.

1.16. **"Curriculum"** means various components / courses studied in each programme that provide appropriate outcomes (knowledge, skills and attitude/ behavior) in the chosen programme of study.

1.17. **"Academic Council"** The committee includes Principal, Dean (Academic), CoE, HoDs, University Nominees, Academic Experts, Industry Experts, Member Secretary and Senior faculty members. This committee is

responsible for framing of academic policies, approval of regulations curriculum and syllabi etc.

1.18. **"Department Advisory Board"** The board includes HoD, Faculty members of the department, Alumni, Experts from other reputed institutions and industries. They review the progress of the department activities and recommend new or revised targets. The board finds the gap between curriculum and industry requirements based on the feedback received from the stakeholders and suggests for suitable corrective actions.

1.19. **"Standing Committee"** consists of Principal, Dean (Academic), Controller of Examination and all the Heads of the Department for smooth functioning of the institution.

1.20. **"Board of Studies (BoS)"** The committee includes HoD, all faculty members of the department, University Nominees, Alumni, Experts from other reputed institutions and industries. Its function includes reviewing and updating curriculum, syllabi from time to time, introducing new courses of study, recommending panel of examiners.

1.21. **"Continuous Assessment"** means evaluation of the students' progress and performance in a course during a semester through various activities such as tests, assignments, quizzes, presentations, practical works, case studies, technology enabled services to community, etc.,

1.22. **"End Semester Examination (ESE)"** means an examination conducted for a course at the end of the semester to evaluate the student's overall learning achievements.

1.23. **"Grade"** means the letter grade (O, A+, A, B+, B and C) based on the marks obtained by a student in a particular course and the range of marks.

1.24. **"Grade point"** means a numerical value (0 to 10) allocated to each letter Grade.

1.25. **"GPA"** means Grade Point Average

1.26. **"CGPA":** It is a measure of overall cumulative performance of a student overall semesters. The CGPA is the ratio of total credit points secured by a student in various courses in all semesters and the sum of the total credits of all courses in all the semesters. It is expressed up to two decimals.

1.27. **"Head of the Institution (HI)"** means the Principal of the College / Institution.

1.28. **"Dean (Academic)"** means the authority of the Institution who is responsible for all academic activities of various programmes and implementation of relevant rules of these Regulations pertaining to the Academic Programmes.

1.29. **"Head of the Department (HoD)"** means Head of the Department concerned.

1.30. **"Controller of Examinations (CoE)"** means the authority of the Institution who is responsible for all the Examination activities conducted by the institution under the autonomous scheme.

1.31. **"Industry"** means the company/organization with which MoU has been signed for various curricular and co-curricular activities.

1.32. **"Capstone Model"** means the components of internal assessment.

1.33. **"CIA"** means Continuous Internal Assessment.

2. ADMISSION REQUIREMENTS

2.1. First Semester Admission

The candidates seeking admission to the first semester of the eight semester Degree of Bachelor of Engineering (B.E) / Bachelor of Technology (B.Tech) programme should have passed the Higher Secondary Examination (10 + 2) in the academic stream with Mathematics, Physics and Chemistry as three of the four subjects of study under Part-III subjects of the study conducted by the Government of Tamil Nadu or any examination of any other University or authority accepted by the Syndicate of Anna University as equivalent thereto.

(OR)

Should have passed the Higher Secondary Examination of Vocational stream (Vocational groups in Engineering / Technology) as prescribed by the Government of Tamil Nadu.

2.2. Lateral Entry Admission

According to the guidelines established by the Government of Tamil Nadu, persons who hold the Diploma in Engineering or Technology recognized by the State Board of Technical Education, Tamil Nadu, or its equivalent, can apply for Lateral entry admission to the third semester of B.E. / B.Tech.

(OR)

Candidates may apply for lateral entry admission to the third semester of B.E. / B.Tech if they hold a B.Sc. degree (10+2+3 stream) from an accredited university with mathematics as one of the topics at the B.Sc. level. According to the college's guidelines, these students must complete two more engineering courses in the third and fourth semesters.

2.3. The candidate should also satisfy other eligibility rules as prescribed by the Anna University and Directorate of Technical Education, Government of Tamil Nadu, Chennai, from time to time.

2.4. **Admissions under NRI / Foreign Nationals (FN) / Overseas Citizens of India (OCI) / Persons of Indian Origin (PIO) / Children of Indian Workers in Gulf Countries (CIWGC)**

Candidates must have passed the qualifying examination i.e (10+2 pattern) of Indian system or its equivalent (completed 12 years of school education). Candidates must have studied in English Medium. The candidate should have passed in all subjects and scored a minimum average of 45% in Mathematics, Physics and Chemistry put together in qualifying examinations. For NRI candidates the eligibility criteria shall be as per the norms of Directorate of Technical Education, Government of Tamil Nadu. The list of qualifying examinations for the candidates in the category of Foreign Nationals (FN) / Overseas Citizens of India (OCI) / Persons of Indian Origin (PIO) / Children of Indian Workers in Gulf Countries (CIWGC) is given in **Annexure 1.**

3. UG PROGRAMMES OFFERED

The following UG programmes of study, approved by Anna University, Chennai and All India Council for Technical Education, New Delhi, offered by the College are shown in Table-1.

TABLE – 1 Programmes Offered

Degree	Programme
B. E.	Biomedical Engineering Computer Science and Engineering Electrical and Electronics Engineering Electronics and Communication Engineering Mechanical Engineering
B.Tech.	Agricultural Engineering Artificial Intelligence and Data Science Information Technology

4. STRUCTURE OF THE PROGRAMMES

4.1 Curriculum and Syllabi

The Curriculum and Syllabi under Regulations 2024 are implemented by considering the guidelines of Anna University, Chennai, the recommendations of AICTE, New Delhi and UGC, New Delhi with a view of achieving excellence in the quality of education by keeping the requirements of enhancing the employability Skill and producing capable engineers for the benefit of Industry, Society and Nation as a whole. Further, the structure of the programme is designed by keeping in mind, the Outcome-Based Education (OBE) and Choice Based Credit System (CBCS).The course content of each course is fixed in accordance with the Programme Educational Objectives (PEOs), Programme Outcomes (POs), Programme Specific Outcomes (PSOs) and Course Outcomes (COs).

4.2 Categorization of Courses

Every Programme shall have a curriculum with syllabi comprising of both theory and practical courses in each semester that have been approved by the respective Board of Studies and Academic Council of the College. The courses are categorized as follows:

4.2.1. **Humanities, Social Sciences and Management courses (HSMC)** include English, Communication skills, Employability Skills, Engineering Ethics Value Education, Economics and Management courses.

4.2.2. **Basic Sciences courses (BSC)** include Mathematics, Physics, Chemistry, Material Science, Biology and further specialization in these courses.

4.2.3. **Engineering Sciences courses (ESC)** include, Engineering Graphics, Engineering practices, Basics of Electrical and Electronics, Civil and Mechanical Engineering, Computer Programming, etc.

4.2.4. **Professional Core courses (PCC)** include the core courses relevant to the chosen branch.

4.2.5. **Professional Elective courses (PEC)** are the courses which can be chosen from a larger pool of courses and are very specific / specialized / advanced / supportive to the discipline which nurtures the students' proficiency in the discipline.

4.2.6. **Open Elective course (OEC):** A student has to choose Open Elective courses from the open electives offered by any B.E. / B.Tech. programme other than his / her own programme with an intention to expose the students to interdisciplinary / multidisciplinary / transdisciplinary fields. The open elective courses on offer are subject to availability of faculty members, time table slot, class rooms and minimum class strength specified from time to time.

4.2.7. **Employment Enhancement Courses (EEC)** Knowledge and Skill Enhancement Courses are offered to the candidates to promote and enhance their technical and skill-based knowledge to increase their employability. EEC include the courses such as Aptitude Essentials, Aptitude Excellence, Innovation through Design Thinking, Product Formulation and Review, Modeling projects, Simulation Using Modern tool, Product Development Practice, Internship, Mini Project, Capstone Project / Industry Practice.

4.2.8. **Mandatory Courses (MC)** includes the courses that are to be completed by every student; these are courses useful for a student's career. These courses will not be included in the CGPA calculation and in the classification of the Degree. Only a PASS is indicated in the Statement of Grades. The details of assessment for the Mandatory

Courses are given in Annexure 2

4.2.9. **Project Work** (PRW) promotes project based learning, team work and innovation culture among the students. Project work shall consist of Project work, Multi-disciplinary Project Work, Product Development Project, Creative Thinking & Innovation Project and Survey Camp etc.

4.2.10. **Audit Courses (AC):** The student may optionally study audit courses prescribed by the College and they will be mentioned in the Grade Statement. However, they will not be considered for computation of CGPA.

4.2.11. **Self-study courses (SSC):** The Department may offer a course which has not been prescribed in the regular curriculum as a self-study course. The syllabus for a self-study course should have been approved by the Chairperson of the concerned BoS. The Students shall study such courses on their own under the guidance of a member of the faculty with no formal lectures need be delivered. Evaluation of suchcourses shall be similar to that of a course in the curriculum. They will be mentioned in the Grade Statement. However, they will not be considered for computation of CGPA.

4.2.12. **Industry Oriented Courses (IOC)** are offered to allow students to expand their knowledge and skills in niche areas and meet the ever changing arising in the relevant industries, from time to time.

4.2.13. The distribution of credits as per the categorization of courses is given in the respective department curriculum and a student has to earn a minimum number of 163 credits in the respective programme for the award of degree.

4.2.14. **Medium of instruction**, examinations and project report shall be in English except certain specified courses.

4.3 Personality and Character Development

All students shall enroll, on admission, in any one of the personality and character development programmes NSS/NSO/YRC/UBA and undergo training / conduct activities for about 80 hours and attend a camp of about

seven days. The training shall include classes on hygiene and health awareness and also training in first-aid. Alternately, activities of science, literature and arts also help for personality and character development. So, students shall conduct and participate actively in Science club/Literary Forum/Fine Arts activities for 80 hours and participate in at least ONE event.

- **National Service Scheme**(NSS) will have social service activities in and around the College / Institution.
- **National Sports Organization** (NSO) will have Sports, Games, Drills and Physical exercises.
- **Youth Red Cross (YRC) / Unnat Bharat Abhiyan (UBA)** will have activities related to social services in and around College/Institution. While the training activities will normally be during weekends, the camp will normally be during vacation period.
- **Science Club** shall organize activities of popularization of science and scientific temper through activities related to astronomy, works of great scientists from India and abroad, observing National Science Day, etc.
- **Fine Arts Club** like music, painting and documentary films with social themes shall be encouraged. Students who enroll and take active participation in anyone of the above activities for 80 hours and participate at least one event/programme will be given a certificate by the Head of the Institution and the copy of the same shall be forwarded to the Controller of Examinations for the purpose of record and scrutiny.
- **Literary Club** like ‘Tamil Ilakkiya Mandram’ shall be formed, which shall organize colorful literary events to propagate good humanist values, morals and ethics reflected in the literature.
- **Research Club** encourages the students and faculty members for research paper presentations, collaborative projects and hands-on workshops, publishing journals and inter college competitions.
- **Language Club** will have the activities related to creative writing, vlogging, vocab quiz and group discussion

- **Quiz Clubs** offers a variety of activities to help students improve their knowledge, critical thinking skills, and communication skills. It provides networking opportunities and collaborative works to identify the talented students.

Fees shall not be charged for any of the above activities. Every student shall put in a minimum of 80% attendance in the events of the club. However, for valid reasons, the Principal may permit a student to complete this requirement before the completion of Programme. Students participate at least one event/programme will be given a certificate by the Head of the Institution and the copy of the same shall be forwarded to the Controller of Examinations for the purpose of record and scrutiny.

4.4 Number of courses per semester

Each semester curriculum shall normally have a blend of lecture courses not exceeding seven for Theory courses and Laboratory integrated theory courses and four for Employability Enhancement Course(s) and Laboratory Courses. However, the total number of courses per semester shall not exceed 10. Each Course shall have credits assigned as per clause 4.5.

4.5 Credit Assignment

Each course is categorized and assigned a certain number of credits in the curriculum. The categorization of courses with number of credits and minimum contact period per week is shown in Table-2 and the number of credits for the contact period is shown in Table-3.

TABLE – 2 Allotment of Credits

Course Category	No. of Credits (C)	Minimum contact period per week		
		L	T	P
Theory	3	3	0	0
Theory with Tutorial	4	3	1	0
Theory with practical combined	4	3	0	2
Practical	1	0	0	2

TABLE – 3 Credit Allocation for Contact Period

Contact Period	Credits
One Lecture (L) period	1
One Tutorial (T) period	1
One Laboratory (1) Period (also for EEC courses like Simulation Using Modern tool / Project / Seminar / Case study etc.,)	0.5

4.6 Induction Programme

The students immediately after admission should undergo a mandatory three weeks induction programme comprising physical activities, yoga, creative arts, universal human values, proficiency modules, motivational and career guidance lectures by eminent experts, visits to local areas and familiarization with the department/programme and training on creativity and innovation.

4.7 Industrial Training / Internship

Each student has to undergo internship in an industry or at the institution. Two weeks of internship at industry/institution is equivalent to 1 credit. The internship report will be evaluated by a team of faculty members nominated by the Head of the Department for awarding the grade. Based on the recommendation by the team, the student will be awarded grade and credit and the results will be sent to The Controller of Examinations to indicate in the Mark Sheet. Each student has to earn 3 credits (6 Weeks) through internship in order to be eligible for the award of the Degree. However, it will not be considered for GPA and CGPA calculations. In addition to the minimum requirement, of 3 credits, if a student earns 3 more credits (6 Weeks of internship in one organization), the student may drop one Programme Elective course. However, if the number of credits earned is 1 or 2, these credits shall not be considered for exemption from Programme Elective Category.

4.8 Industrial Visit

Every student is expected to go for at least one Industrial Visit to get experiential learning every year starting from the second year of the Programme. The Heads of Departments shall ensure that necessary arrangements are made in this regard.

4.9 Value Added Courses (VAC):

- Students have the option of registering for Value Added Course(s) (VAC) and the credits earned after successful completion of VAC shall be featured in the Grade sheet. However, credits earned (if any) will not be considered for computation of CGPA. Students shall be allowed to take these courses offered in other Departments also, but with the permission of the Head of the Department of student and Head of the Department offering the course.

- The details of the syllabus, timetable and faculty shall be sent to the head of the Institution with the recommendation of the respective Departmental Advisory Board (DAB) for approval. The courses once approved by the Institute represented by any Department shall be notified in the College website and these courses can also be offered by other institute Departments.

- The result(s) of VAC shall be printed in grade sheet, only upon successful completion of the course. However, the credits earned shall not be considered for computation of CGPA.

4.10 Online Courses (OC):

Students may be permitted to credit a maximum of two online courses, subject to a maximum of six credits, with the approval of the **Head of the Institution, in lieu of Open Elective / Professional Elective courses. The Head of the Institution shall form a three member committee with members as HOD and a faculty member from the Department of the student, HOD of any other branch of the Institution to ensure that the student has not studied such courses and would not repeat it again as Professional Core/Professional Elective/Open Elective courses.** Suitable online courses shall be chosen from the SWAYAM platform, Udemy, Coursea etc.

5. Academic Calendar

The dates of all academic activities including those of course registration, first and the last days of classes, examinations, supplementary

examinations and vacation are published in the Academic Calendar every year. The academic calendar for each year shall be available in the college website.

6.Duration of the Programme

6.1.1. Duration of the programme is FOUR years (Eight Semesters) for a student admitted to a programme with qualifications as prescribed in Clause 2.1 of the regulations. The student shall successfully complete all the requirements as per the curriculum and regulations of the programme within the above said duration of 'FOUR' years.

6.1.2. Duration of the programme is THREE years (Six Semesters) for a student admitted directly to second year (Lateral entry) with qualifications as prescribed in Clause 2.2 of the regulations. The student shall successfully complete all the requirements as per the curriculum and regulations of the programme within the above-said duration of 'THREE' years.

6.1.3. Above mentioned period in Clause 6.1.1 & Clause 6.1.2 shall be extended for 'ONE' more year to a student, who is permitted to avail the authorized break of study or opt for dropping of courses, as per the regulations.

6.1.4. 'Programme Completion' means that a student shall undergo all the courses as prescribed in the respective curriculum and become eligible for appearing in the end semester examination of all the courses within the period as given in Clauses 6.1.1, 6.1.2 & 6.1.3. Eligibility here refers to issuance of hall ticket to the student for appearing in the end semester examination.

6.1.5. After programme completion if there are any arrear course(s), a student shall be permitted to successfully complete the same, within 'SEVEN' years (students admitted under Clause 2.1) and 'SIX' years (students admitted under Clause 2.2). This maximum duration for successful completion shall be reckoned from the date of original admission to the programme.

6.1.6. In case of a student prevented from appearing in the end semester examination in all the courses of a semester due to shortage of attendance, the period of prevention shall be treated as a break in the programme. In

such a case, total duration for completing the programme including all such prevention periods shall not exceed the above-said 'SEVEN' years (students admitted under Clause 2.1) and 'SIX' years (students admitted under Clause 2.2).

7.COURSE ENROLLMENT REGISTRATION

7.1. The information on the list of all the courses offered in every department specifying the credits, the prerequisites, a brief description of syllabus or list of topics, the instructor who is offering the course shall be made available in the college website.

7.2. Each student, on admission shall be assigned to a Tutor/Mentor (vide clause 9.0) who shall advise her/him about the academic programs and counsel on the choice of courses considering the academic background and student's career objectives. With the advice and consent of the Tutor / mentor, the student shall register for a set of courses he/she plans to take up for the Semester.

7.3. Every student shall enroll for the courses of the succeeding semester during the last week of the current semester. However, the student shall confirm the enrollment by registering for the courses within first five working days after the commencement of the concerned semester. Further, the student has to register for the end semester examination by paying the prescribed fee for that particular course. If he/she has not registered for the end semester examination in the semester concerned, then he/she has to re-register for the courses in the subsequent semesters and satisfy the requirements as per clauses 7.11.2.

7.4. If a student is prevented from writing End Semester Examination (ESE) of a course due to lack of attendance, the student has to register for that course again, when offered next, attend the classes and fulfill the attendance requirements as per clause 8.

7.5. If the theory course in which the student has been prevented from writing end semester examination due to lack of attendance is a programme elective course or an open elective, then the student may register for the same or any other Programme elective or open elective course respectively

in the subsequent semesters during the minimum period of study. However, if the students are unable to complete the courses within the minimum said duration, they shall register their courses in the subsequent academic year irrespective of Odd or Even semester.

7.6. The maximum number of credits that can be registered in a semester is 36. However, this does not include the number of Re-appearance (RA) and Withdrawal (W) courses registered by the student for the appearance of Examination.

7.7. In any department, the preference for registration shall be given to the students of that department for whom the course is a programme core course or programme elective course.

7.8. The registration for any course shall be on first come first served basis, provided the student fulfills prerequisites for that course if any. Every effort shall be made by the Department/Centre to accommodate as many students as possible.

7.9. No course shall be offered by a department unless a minimum of 5 students are registered for that course. However, this restriction is not applicable for the students who register for a course after 4 years duration.

7.10. After registering for courses for the semester, a student shall attend the classes, satisfy the attendance requirement as per clause 8, earn Continuous Assessment marks and appear for the End Semester Examinations.

7.11. **Flexibility to Add or Drop a course**

A student has to earn the minimum total number of credits specified in the curriculum of the respective Programme in order to become eligible to obtain the degree. However, if the student wishes, the student is permitted to earn more than the total number of credits prescribed in the curriculum of the student's programme.

7.11.1. From the fifth to seventh semesters, the student has the option of registering for additional courses or dropping existing courses. Total number of credits of such courses shall not exceed 6 per semester.

The student is permitted to drop the course(s) within 30 days of the commencement of the academic schedule. However, the student shall register for a **maximum of 36 credits** and a minimum of 12 credits.

7.11.2. If a student earns more than the total credit as specified in the curriculum of the programme by using this add on flexibility, the courses with higher grades shall be considered for the calculation of CGPA. Remaining courses shall be printed in the mark sheet, however, they will not be considered for calculation of CGPA.

7.11.3. A student can register a maximum of 2 one-credit courses per semester for the replacement of programme elective and open elective courses.

7.11.4. The student shall register for the Capstone project / Industry Practice in the VIII semester only.

7.11.5. The student shall register for the Product Development practice in the VI semester.

7.11.6. The student shall register and complete their courses within the minimum duration of the degree programme as per clause 6.1.1 & 6.1.2. However, if a student is unable to register the courses within the said duration, they shall register their courses in the subsequent academic year irrespective of Odd or Even semester.

7.12. **Reappearance for arrear examinations**

7.12.1. If a student fails in a theory course, the student shall appear for the End Semester Examinations in the subsequent semester(s). The registration of reappearance courses in the subsequent semester is mandatory.

7.12.2. This applies to all theory courses including ‘Theory with Laboratory component’ courses. The passing requirements for such courses shall be as given in clause 16.

7.12.3. If a student fails in a professional elective or an open elective, he/she may reappear for the End Semester Examination of the same course(s) in the subsequent semester(s). Instead, if the student wishes to register

for different professional elective or open elective course(s) in the subsequent semester(s), the student should attend the classes, earn continuous assessment marks and appear for End Semester Examination. However, such change of elective courses is permitted only once during the entire duration of study.

7.12.4. A student who fails in any Laboratory Course / Project work / Seminar and any other EEC course shall reappear for End Semester Examination in the subsequent semester(s). For all theory, laboratory and EEC courses, the passing requirements are as given in clause 16.

7.12.5. If a student is prevented from writing End Semester Examination of a course due to lack of attendance, the student has to register for that course again, when offered next, attend the classes and fulfill the attendance requirements as per clause 8. If the course, in which the student has “lack of attendance”, is a Professional Elective or Open Elective, the student may register for the same or any other Professional Elective or Open Elective course(s) in the subsequent semester(s) and appear in the examination as per Clause 7.10.

7.12.6. An Elective course shall be offered by any department for a minimum 20 students registering for the course. However, if the student’s strength is less than one division, a minimum of 50% of students shall register for an Elective course.

7.13. **Choice of Professional Elective Courses**

The professional Elective Courses are listed in the Curriculum in Table format as verticals (Specialization groups). A student can choose all the Professional Elective Courses either from one of the verticals or a combination of courses from all verticals in a semester.

However, students irrespective of enrolling for additional courses for B.E. / B. Tech. (Hons.) are not permitted to choose more than one course from a row. Students are permitted to enroll more than one elective course from the same vertical in a semester. In the subsequent semesters students are permitted to enroll one more course in a row, provided if he/she has cleared the earlier course of the same row.

For a professional elective course and open elective course, minimum number of students enrolment permitted shall be 10. However, the minimum number is not applicable for students enrolling B.E. / B. Tech. (Hons) and B.E. / B. Tech. Minor. For the offer of each professional elective at least two choices shall be offered.

8. ATTENDANCE REQUIREMENTS FOR COMPLETION OF THE SEMESTER

A student who has fulfilled the following conditions shall be deemed to have satisfied the attendance requirements for appearing for End Semester Examination of a particular course.

8.1. Every student is expected to attend all the periods and earn 100% attendance. However, a student shall secure not less than 75% attendance overall taking into account the number of periods required for that semester courses as specified in the curriculum.

8.2. If a student, secures overall attendance between 65% and 74% in the current semester due to medical reasons (prolonged hospitalization / accident / specific illness) or participation in Institution / University / State/ National / International level extra and co-curricular activities, with prior permission from the Head of the Department, shall be permitted to appear for the current semester examinations subject to the condition that the student shall submit the medical certificate / participation certificate attested by the Head of the Department (along with Condonation form). Such certificates along with the Condonation forms shall be forwarded to the Controller of Examinations for verification and permission to attend the examinations. However during the entire programme of study, a student can avail such Condonation in any two semesters only.

8.3. A student shall normally be permitted to appear for End Semester Examination of the course(s) if the student has satisfied the attendance requirements (vide Clause 7.1) and has registered for examination in those courses of that semester by paying the prescribed fee.

8.4. Students who do not satisfy Clause 8.1 and 8.2, and those who secure less than 65% attendance will not be permitted to write the End-Semester Examinations of that semester and are not permitted to go to next semester.

They are required to repeat the incomplete semester in the next academic year.

8.5. Student's progress, performance and conduct are satisfactory.

9. MENTOR / FACULTY ADVISOR (Class Co-ordinator)

To help the students in planning their courses of study and for general advice on the academic programme, the Head of the Department will nominate a teacher of the Department who shall function as Faculty Advisor for the students throughout their period of study. The Faculty Advisor shall advise the students in registering of courses, monitor their attendance and progress and counsel them periodically. If necessary, the Faculty Advisor may also discuss with or inform the parents about the progress / performance of the student concerned.

The responsibilities for the faculty advisor are:

- To inform the students about the various facilities and activities available to enhance the student's curricular and co-curricular activities.
- To guide the student enrollment and registration of the courses.
- To monitor the academic and general performance of the students including attendance and to counsel them accordingly and
- Any other requirement which may arise from time to time.

10. COURSE COMMITTEES

10.1. CLASS COMMITTEE

For all the courses taught, prescribed in the curriculum, Class Committee meeting shall be convened thrice in a semester (first meeting within 15 days from the commencement of the semester and other two meetings at equal interval after the first meeting) comprising members of the faculty handling all the courses, mentors and six student representatives from the class.

One of the members of the faculty (preferably not handling any courses to that class), nominated by the Head of the Department, shall coordinate the activities of the Committee. During these meetings, the student members shall meaningfully

interact and express their opinions and suggestions of all the students to improve the effectiveness of the teaching-learning process. It is the responsibility of the student representatives to convey the proceedings of these meetings to all other students.

10.2. COURSE COMMITTEE FOR COMMON COURSES

10.2.1. A theory course handled by more than one faculty including the discipline with multiple divisions (greater than or equal to 2) shall have a “Common Course Committee” comprising of all members of faculty teaching that course with one of the members as the Course Coordinator, nominated by the Head of the Institution (Head of the Department in the case of multiple divisions of a discipline) and student representatives (one per specialization or division) registered for that course in the current semester. First meeting of the Common Course Committee shall be held within fifteen days from the date of commencement of the semester. Four subsequent meetings in a semester may be held at suitable intervals.

10.2.2. In addition, Common Course Committee shall meet the student representatives of each class. During these meetings, the student members shall meaningfully interact and express their opinions and suggestions of all the students to improve the effectiveness of the teaching- learning process. It is the responsibility of the student representatives to convey the proceedings of these meetings to all the students

10.3. Overall Monitoring Committee

Overall Monitoring Committee (OMC) is constituted to monitor all academic activities, progress and status of the students of the semester concerned. OMC comprises of (i) Head of the Department (ii) All mentors of the students. The OMC will meet periodically to discuss all academic activities and progress of the students. OMC can also invite some students for any of the committee meetings if necessary.

11. SYSTEM OF EXAMINATION

11.1. All B.E./B.Tech. Programmes consist of end semester examinations in all Theory and Laboratory Courses, Project Work, etc. as mentioned in the

curriculum. Appearance in examinations is mandatory for all the courses.

11.2. Performance in each course is evaluated based on (1) Continuous Internal Assessment (CIA) throughout the semester and (2) End Semester Examination (ESE) at the end of the semester. All the courses are evaluated for the specified maximum marks with the apportionment of marks for CIA and ESE components as shown in Table-4.

TABLE – 4 ASSESSMENT

S. No.	Category of Course	Maximum Marks	Apportionment of	
			CIA Marks	ESE Marks
1.	Theory Courses	100	40	60
2.	Theory and practical combined courses	100	50	50
3.	Practical Courses	100	60	40
4.	Project Work / Mini ProjectWork	100	40	60
5.	In-plant Training	100	100	-
6.	Internship	100	100	-
7.	All other courses	100	100	-

11.3. Industrial Training and Seminar shall carry 100 marks and shall be evaluated through internal assessment only.

11.4. The End Semester Examination (theory and practical) of 3 hours duration shall ordinarily be conducted between November and December during the odd semesters and between April and May during the even semesters.

11.5. The End Semester Examination for Project Work shall consist of evaluation of the final report submitted by the student or students of the project group (of not exceeding 4 students) by an external examiner and an internal examiner, followed by a viva voce examination conducted separately for each student by a committee consisting of the external examiner, the supervisor of the project group and an internal examiner.

11.6. For the End Semester Examination of practical courses including Project Work, the internal and external examiners shall be appointed by the Controller of Examinations

11.7. If a student indulges in malpractice in any of the End Semester Examinations, he/she shall be liable for punitive action as prescribed by the Enquiry Team constituted by the Head of the Institution from time to time.

12. PROCEDURE FOR AWARDING MARKS FOR CONTINUOUS ASSESSMENT

For all theory, practical courses, theory courses with practical combined and project work the continuous assessment shall be awarded as per the procedure given below:

12.1. Theory Courses

Evaluation shall be through continuous assessments and end semester examination. Continuous assessment shall have the weightage of 40% & End semester examination shall have the weightage of 60%.

12.1.1. Continuous assessments comprise of two tests and activities. For each test, there shall be a minimum of ONE 'activity' which includes tutorials, assignments, group assignments, quizzes, etc. Weightage for test is 80% and for activity, it is 20%, as provided in Table 5. Average of marks obtained in two assessments, shall be taken as Continuous Assessment marks for the particular course.

TABLE – 5 Weightage for Continuous Assessment in Theory (T) Courses

Assessment – I	Assessment – II	Activities	Total
40%	40%	20%	100%

12.1.2. End semester examination shall be conducted for 100 marks for duration of THREE hours.

In case a student misses the internal assessment due to medical reasons(hospitalization / accident / specific illness) or due to participation in the College / University / State / National / International level Sports events with prior permission from the Chairman, Sports Board a Reassessment may be conducted before the end of the semester after getting approval for the Head of the Institution.

12.2. **Laboratory courses**

Evaluation shall be through continuous assessments and end semester examination. Continuous assessment shall have the weightage of 60% & End semester examination shall have the weightage of 40%.

12.2.i. Continuous assessments will be based on turn-to-turn supervision of the student's work, submission of observation notebooks, etc. and it has the weightage of 75%. Performance in laboratory practical test (Mid-term) shall have the weightage of 25% as given in Table 6. Rubrics for each laboratory course shall be arrived in the common course committee at the beginning of the semester.

TABLE – 6 Weightage for Continuous Assessment in laboratory (L) Courses

Evaluation of Student's work, Observation, Record, etc.	Model Examination
75%	25%

12.2.ii. There shall be one End semester examination for laboratory course conducted for 100 marks and this examination shall be conducted preferably before the commencement of theory examinations.

**The total marks shall be converted into 60 marks for Internal Assessment.*

Rubric will be framed by the course coordinator and duly approved by the Head of the Department.

12.3. **Theory Courses with Practical Combined / Practical course with Theory Combined**

Evaluation shall be through continuous assessments and end semester examination. Continuous assessment shall have the weightage of 50% & End semester examination shall have the weightage of 50%.

12.3.i. Continuous assessments comprise of two tests for theory and one test for practical. For each theory test, there shall be a minimum of

ONE 'activity' which includes assignments, group assignments, quizzes, etc. The assessment of laboratory practical component will be based on turn-to-turn supervision of the student's work, performance in laboratory practical test (Mid-term). Rubrics for laboratory practical component shall be arrived in the common course committee at the beginning of the semester. Details of weightage for theory and practical component with respect to credits are presented in Table 7.

TABLE – 7 Weightage for Continuous Assessment in LIT Courses

S. No.	L	T	P	C	Continuous Assessment	
					Theory & Activities*	Practical
1.	1	0	2	2	25%	25%
2.	1	0	4	3	15%	35%
3.	2	0	2	3	35%	15%
4.	3	0	2	4	35%	15%
5.	2	0	4	4	25%	25%
6.	3	0	3	4.5	35%	15%
7.	3	0	4	5	35%	15%

* 5% shall be allotted to the activities.

12.4. **Assessment for Project Work**

12.4.1 Project work / Mini Project work is allotted to a single student or to a group of students not exceeding 4 per group.

12.4.2 Project / Mini Project Work shall be carried out under the supervision of a "qualified teacher" in the Department concerned. In this context "qualified teacher" means thefaculty member possessing (i) PG degree or (ii) Ph.D. degree.

12.4.3 The Project / Mini Project Work carried out in an industry / academic/ research institution during internship shall be jointly supervised by a supervisor of the department and an expert as a joint supervisor from the organization and the student shall be instructed to meet the supervisor periodically and to attend the review committee meetings for evaluating the progress. The review meetings, if necessary, may also be arranged in online mode with prior approval from the Principal and suitable record of the meetings shall be maintained.

12.4.4 The HOD shall constitute a review committee for Project / Mini Project Work. The review committee consists of the supervisor, an expert from the Department and project coordinator from the Department. If the project coordinator/expert member happens to be the supervisor, then an additional member shall be nominated. In the case of Industrial Project, the review committee shall have the supervisor, an expert from the industry and the project coordinator from the Department.

12.4.5 There shall be three reviews during the semester by the review committee. The student shall make presentation on the progress made by him / her before the committee. The total marks obtained in the three reviews shall be reduced for 60 marks and rounded to the nearest integer.

12.4.6 The project report shall carry a maximum of 20 marks. The project report shall be submitted as per the approved guidelines. Same marks shall be awarded to every student within the project group for the project report. The viva-voce examination shall carry 20 marks. Marks are awarded to each student of the project group based on the individual performance in the viva-voce examination.

12.4.7 The student(s) is expected to submit the project report on or before last working day of the semester. The End Semester Examinations for project work shall consists of evaluation of the final project report submitted by the students or students of the project group by an external examiner followed by a viva-voce examination conducted separately for each student by a committee consisting of the external examiner, the guide of the project group and an internal examiner. The Controller of Examinations shall appoint Internal and External Examiners for the End Semester Examinations of the Project Work.

TABLE – 8: Weightage for Continuous Assessment & End-Semester Examination for Project Work

Continuous Assessment (60 Marks)			End Semester Examinations (40 Marks)			
ReviewI	ReviewII	Review III	Project Thesis Report Evaluation		Viva-Voce Examination	
			Guide	External examiner	External examiner	Internal Examiner
10%	20%	30%	10%	10%	10%	10%

12.4.8 If a student fails to submit the project report on or before the specified submission date or the student has submitted the project report but did not appear for the viva- voce examination, it will be considered as fail in the Project Work and the student shall re-register for the same in the subsequent semester.

12.4.9 If the student fails to secure 50% of the semester examination marks in the project work, he / she have to repeat the project work during the next semester. A fresh viva voce examination may be conducted at the end of the next semester.

12.4.10 A copy of the approved project report, after the successful completion of viva - voce examination, shall be kept in the department library.

12.5. **Assessment for 'Internship cum Project Work'**

A student or a group of students (not exceeding four) who are selected for the internship, shall work in a project in the industry concerned that mandatorily requires an internal guide & external guide. On receipt of the intimation letter / Email regarding the internship, the Head of the Department concerned shall nominate a Guide and Co-Guide (if required), for smooth progression of the works.

12.5.1. Internal Guide is a faculty from the department of the student. An internal co-guide from other department may also be nominated by the Head of the Department concerned, based on the nature of the project work, if required.

12.5.2. An external guide shall be identified at the time of offering the internship to the students by the industry concerned and the external guide has to interact with the internal guide / co-guide at regular intervals for formulation, monitoring and completion of the project work. External guide shall be a review committee member for evaluating the progress of the work and performance of the student(s). Bona fide certificate for the completion of the 'Internship cum Project Work' shall be approved by both internal guide(s) and external guide.

12.5.3. Any confidential / Intellectual Property Rights matters related to the industry concerned, shall be discussed by the external guide from the industry, at the beginning of the internship cum project work with the student(s) and the internal guide(s), in order to arrive at the problem definition & formulation of objective(s) of the Internship cum Project Work.

12.5.4. For undergoing 'Internship cum Project Work' in hassle-free manner and also in the large academic interest of the students and also to maintain the quality of the project work, sufficient information and data shall be provided by the industry concerned to the student / group of students and the same shall be ensured by the internal guide(s) at the beginning and at regular intervals, for successful completion of the 'Internship cum Project Work'.

12.5.5. Students who are undergoing 'Internship cum Project Work' in the industries located in the vicinity of Chennai (60 km from the University) shall attend the review in physical mode. Whereas those students who undergo 'Internship cum Project Work' in the industries located beyond 60 km from the University shall be allowed to attend the review, either in physical or online mode. Importantly, the external guide shall attend the review, either through online or physical mode.

12.5.6. Evaluation of 'Internship cum Project Work' is as per Clause 12.4.

12.6. Assessment for Internship / Industrial training

A student shall undergo 'Internship / Industrial Training' for a period as stipulated in the respective curriculum of the Programme. The industry in which the student proposes to undergo Internship / Industrial Training, shall involve in Design / Fabrication / Production / Manufacturing / Testing / Research & Development / Services, etc. in alignment with the fields related to the programme. After completion of the Internship / Industrial Training, the student shall submit a report, describing the observations, skills acquired, usefulness, etc., along with the attendance certificate issued by the industry concerned. Evaluation shall be by the committee comprising of three faculty members, and a representative from the industry concerned (preferably Unit Head) constituted by the respective Head of the Department and one among the faculty shall be nominated as the coordinator. Marks are awarded to the students based on the evaluation with the weightage of 40% for report evaluation, 30% weightage for presentation and 30% weightage for viva-voce examination (out of which 10% marks from the industry concerned), as presented in Table 9. In case of pre-occupation of the industry representative for the conduct of viva-voce examination, then the marks awarded by the representative shall be obtained through email.

TABLE – 9: Weightage for Continuous Assessment of Internship / Industrial Training

Project Report	Presentation	Viva-voce	Total
40%	30%	30%	100%

12.7. Assessment for Mini project

An academic activity as part of the curriculum, a student or a group of students shall undergo the practice of doing theoretical or experimental works for understanding the concepts elaborately. A review committee shall be constituted by the respective Head of the Department, consisting of three faculty members viz. an internal expert, coordinator and respective supervisor for periodical reviews. Performance of the students is assessed by this committee only through continuous assessments, which include three reviews. Weightage is 25% each for the first two reviews and 50% for the

third review as given in Table 10. Student(s) shall submit a report in the prescribed format within the deadline as specified by the Head of the Department. This committee evaluates the project report submitted by the student(s) with a weightage of 30% out of 50% and conduct viva-voce examination with a weightage of 20% out of 50%, in the third review.

TABLE – 10 Weightage for Assessment of Mini Project

Review I	Review II	Review III	
		Report	Viva-Voce Examination
25%	25%	30%	20%

12.8. **Assessment for Seminar Courses**

Students are encouraged to present a topic of their interest, related to the programme and approved by the Course Instructor, for better understanding of concepts, promote critical thinking and enhance the presentation as well as the communication skills of the students. Evaluation shall be only through continuous assessments by the Course Instructor, with equal weightage of the marks obtained in a minimum of three presentations by a student. End Semester Examinations shall not be conducted for this course.

12.9. A course on **'Standards'** is discipline-specific, offered in order to equip the students with essential knowledge on standards with respect to their discipline of study. Performance of the students shall be evaluated only through continuous assessments with minimum of 'TWO' tests. Out of two tests, one test shall be conducted at the mid-term of the semester and second test shall be conducted at the end of the semester. These two tests shall carry equal weightages. End Semester Examinations shall not be conducted for this course.

12.10. **Assessment for Value Added Courses**

The value added course shall carry 100 marks and shall be evaluated through continuous assessments only. Two Assessments shall be conducted during the semester by the Department concerned. The total marks obtained in the tests shall be reduced to 100 marks and rounded to

the nearest integer. A committee consisting of the Head of the Department, staff handling the course and a senior faculty member nominated by the Head of the Institution shall do the evaluation process. The list of students along with the marks and the grades earned shall be forwarded to the Controller of Examinations for appropriate action at least one month before the commencement of End Semester Examinations. The grades earned by the students for Value Added Courses will be recorded in the Grade Sheet, however the same shall not be considered for the computation of CGPA.

12.11. Assessment for Online Courses

Students may be permitted to credit two online courses (which are provided with certificate), subject to maximum of six credits. The online course of 3 credits can be considered instead of one elective course. These online courses shall be chosen from the SWAYAM platform, provided the offering department conducts regular examination and provides marks. The credits earned shall be transferred and the marks earned shall be converted into grades and transferred, provided the student has passed in the examination as per the norms of the offering institution. The details regarding online courses taken up the student and marks/credits earned and the approval for the courses from Head of the Institution shall be sent to the Controller of Examinations, in the subsequent semester(s) along with the details of the elective(s) to be dropped.

The credits earned from approved SWAYAM online courses by a student may be taken into account for the total credits to be earned and CGPA calculation. Mapping of Marks scored in NPTEL course and Credits earned as per Table-11.

Table 11: Duration of the Course and Number of Credits

S. No.	No. of Weeks	No. of Credits
1.	4	1
2.	8	2
3.	12	3
4.	16	4

Table 12: Mapping of Marks scored in NPTEL course and Credits earned

Letter Grade	Marks
O	90 – 100
A+	80 – 89
A	70 – 79
B+	60 – 69
B	50 – 59
C	40 – 49
U	< 40

12.12. **Self-study courses**

A student may be permitted to study a course on his own without attending a class with the approval of the HOD. The member of faculty approved by the Head of the Department shall be responsible for periodic monitoring and evaluation of the course. The course shall be evaluated through Continuous Assessment and Semester Examination. The evaluation methodology shall be the same as that of a theory course.

12.13. **Assessment for Mandatory Courses**

Courses on peripheral subjects in a programme, wherein familiarity is considered mandatory, are included as mandatory courses in the curriculum. A student has to obtain PASS in all Mandatory Courses to qualify for the degree. Mandatory course shall carry 100 marks and shall be evaluated through internal assessment only.

12.14. **OTHER EMPLOYABILITY ENHANCEMENT COURSES**

- The Seminar / Case Study / Mini Project course is to be considered as purely INTERNAL (with 100% internal marks only). Every student is expected to present a minimum of 2 seminars per semester before the evaluation committee and for each seminar, marks can be equally apportioned. The three member committee appointed by the Head of the Institution, consisting of the course coordinator and two experts from the Department, will evaluate the seminar and at the end of the semester, the marks shall be consolidated and taken as the final mark. The evaluation

shall be based on the seminar paper (40%), presentation (40%) and response to the questions asked during presentation (20%).

- The Industrial / Practical Training, Summer Project, Internship, shall carry 100 marks and shall be evaluated through internal assessment only. At the end of Industrial / Practical Training / Internship / Summer Project, the student shall submit an attendance certificate from the organization where he / she had undergone training and a brief report. The evaluation will be made based on this report and a viva-voce Examination, conducted internally by a three member Departmental Committee constituted by the Head of the Institution consisting of the course coordinator and two experts from the Department. The certificates (issued by the organization) submitted by the students shall be attached to the mark list sent by the Head of the Institution to the Controller of Examinations.

- For all the courses under Employability Enhancement Courses Category, except the Project Work, the evaluation shall be done with 100% internal marks.

12.15. Internal marks approved by the Head of the Institution shall be displayed by the respective HODs within 5 days from the last working day.

12.16. **Supplementary Test:** A student becomes eligible to appear for Supplementary Test (s) conducted by the Controller of Examinations, after the Continuous Internal Assessment(s) (CIA), only under the following circumstances: (i) The student was absent for CIA on account of medical reasons (hospitalization / accident / specific illness), or participation in the College / University / State / National / International level Sports events with prior permission from the Head of the Department, and (ii) the student should have registered for the Supplementary Test, through the member of faculty who handles the course or through the respective Head of the Department and got approval by the Head

12.17. **Faculty Attendance and Assessment Record**

Every faculty member is required to maintain an '**ATTENDANCE AND ASSESSMENT RECORD**' which consists of attendance marked in each theory or Laboratory class, the assessment marks and the record of class

work (topics covered), separately for each course . This should be submitted to the Head of the Department periodically (at least three times in a semester) for monitoring the syllabus coverage and the records of assessment marks and attendance. The Head of the Department will affix the signature and date after due verification. At the end of thesemester, the record should be verified by the Head of the Department who shall keep this document in safe custody (for five years). The records of attendance and assessment of both current and previous semesters should be available for inspection.

12.18. **Conduct of Academic Audit**

Every educational institution shall strive for a better performance of the students by conducting the internal assessments as mentioned in Clause 11.

In order to ensure the above, Academic Audit is to be done for every course taught during the semester. For the internal assessments conducted for each course as per details provided in Clause 12, the academic records shall be maintained in the form of documentation for the individual assignments / case study report / report of mini project submitted by each student and assessment test question paper and answer script. Report of industrial training / internship shall also be maintained, if applicable. For laboratory courses students' record shall be maintained. Further, the attendance of all students shall be maintained as a record.

The Head of the Institution shall arrange to conduct the Academic Audit for every course in a semester by forming the respective committees with an external course expert as one of the members drawn from a Technical institution of repute near the institute.

The University or any inspection team appointed by the University may verify the records of Academic Audit report of the courses of both current and previous semesters, as and when required.

13. REQUIREMENTS FOR APPEARING FOR END SEMESTER EXAMINATIONS

A student shall normally be permitted to appear for the University Examinations for all the courses registered in the current semester (vide clause 7) if he/she has satisfied the semester completion requirements (as per clause 8).

Further, examination registration by a student is mandatory for all the courses in the current semester and all arrear(s) course(s) for the University examinations failing which, the student will not be permitted to move to the higher semester.

A student who has already appeared for any course in a semester and passed the examination is not entitled to reappear in the same subject for improvement of grades.

In the case of examination in project work, Mini-projects, Internship project, Industrial project and literature review, no student will be permitted to appear in the project work examination unless he / she has submitted the project report within the prescribed date (under normal circumstances)

A student shall become eligible for appearing in End Semester Examination only after he/she has paid all the fee prescribed by the college before commencement of End Semester Examinations which includes course fee (Including arrear if any), hostel fee, examination fee and any other fee decided on time to time.

14. Procedure for Rejoining the Programme

A student who is required to repeat the study of any semester for want of attendance/progress/conduct or who desires to rejoin the programme after a period of discontinuance or who upon his/her own request is permitted by the authorities to repeat the study of any semester, may join the semester which he/she is eligible or permitted to join, only at the time of its normal commencement for a regular batch of students and after obtaining the approval from Directorate of Technical education and affiliating university. In the case of repeaters, the continuous assessment marks secured earlier in the repeated courses will be discarded. A new register number shall be assigned to the candidate.

15. Procedure for Transfer of Students (Student migration and Credit Transfer)

Normalization of the credits will be carried out in consultation with the Board of Studies of the programme concerned and approved by the Head of the Institution, if a student migrates from other institutions to Indra Ganesan College of Engineering or rejoins from previous regulation to this regulation.

16. PASSING REQUIREMENTS

16.1. A student who secures not less than 50% of total marks prescribed for a course [Internal Assessment + End semester Examinations] with a minimum of 45% of the marks prescribed for the End Semester Examination, shall be declared to have passed the course successfully and earned the prescribed credits for that course, applicable for all registered courses. This is applicable for both theory and practical courses (including project work).

16.1.1 If a student fails to secure a pass in a particular course, i.e., failing to obtain minimum marks, as stated above, it is mandatory that he/she shall reappear for the examination in that course in the subsequent semester(s) whenever the examinations are conducted for that course, till he / she secures a ‘Pass’. If the course is a ‘Theory with Laboratory component’, he/she shall write the End semester examinations for the Theory component only.

16.1.2 Continuous Assessment (CA) marks obtained by the student in the first appearance shall be retained and considered valid for two subsequent attempts. However, from the third attempt (i.e., fourth appearance) onwards, the student shall be declared to have passed the course if he/she secures a minimum of 50% minimum marks in the End Semester Examinations alone and the related Continuous Internal Assessment marks obtained will not be considered or retained thereafter.

16.2. Students’ need to earn a minimum number of 163 credits to qualify for the award of Degree in the various branches of study as prescribed by the respective Boards of Studies and Academic Council.

17. AWARD OF LETTER GRADES

17.1. The assessment shall be based on the performance in the End Semester Examinations and / or Continuous Assessment, carrying marks as specified in Clause 11. Letter Grades (based on Credit Point and Grade Point) are awarded to the students based on the performance in the evaluation process.

17.2. Credit Point is the product of Grade Point and number credits for a course and Grade Point is a numerical weight allotted to each letter grade on a 10-point scale (as specified in the Clause 17.3), while the Letter Grade is an index of the performance of a student in a said course.

17.3. The performance of a student will be reported using Letter Grades, each carrying certain points as detailed below:

Range of total marks (as specified in Clause) / Specific reason	Grade Points	Letter Grade
90 – 100	10	O (Outstanding)
80 – 89	9	A+ (Excellent)
70 – 79	8	A (Very Good)
60 – 69	7	B+ (Good)
50 – 59	6	B (Above Average)
0 – 49	0	RA (Reappearance)
Withdrawal of course	0	W
Shortage of Attendance	0	SA
Withheld		WH

"RA" --- Reappearance

'W" --- Withdrawal from End Semester Examination

"SA" --- Shortage of attendance (clause 7) and hence prevented from writing end semester examination.

17.4. **GPA and CGPA Calculation:**

17.4.1 Two parameters, Grade Point Average (GPA) and Cumulative Grade Point Average (CGPA) are used to evaluate the academic performance

of the student.

17.4.2 The GPA is the weighted average of the grade points obtained in all the courses registered by the student during the semester. For example, if a student passes five courses (Theory / Laboratory / Project / Seminar etc.) in a semester with credits C_1, C_2, C_3, C_4 and C_5 and the students' grade points in these courses are g_1, g_2, g_3, g_4 and g_5 respectively, then GPA of the student is calculated as:

$$GPA = \frac{(C_1 \text{ x } g_1) + (C_2 \text{ x } g_2) + (C_3 \text{ x } g_3) + (C_4 \text{ x } g_4) + (C_5 \text{ x } g_5)}{(C_1 + C_2 + C_3 + C_4 + C_5)}$$

17.4.3 At the end of each semester, the student is given with a semester wise grade sheet. This is a consolidated list of courses registered and grades obtained in that particular semester along with the appropriate GPA calculation.

17.4.4 The CGPA indicates the overall academic performance of a student and is computed to two decimal places in the same manner as the GPA, except that all the courses registered up to and including the latest completed semester. Grades obtained after successful completion of additional courses, which are not mandatory for minimum requirement for award of degree, shall not be considered for computation of CGPA. However, results of these courses will be printed only in the grade sheet.

17.4.5 For the students who have successfully completed the requirements of Honours Degree, grades obtained in the additional courses / Capstone Design Project shall be taken into account for the computation of CGPA.

17.4.6 For the students who have successfully completed the requirements of Minor Degree, grades obtained in these additional courses shall be taken into account for the computation of CGPA.

17.4.7 If the student is absent for the registered course in the End Semester Examination(s) or Viva-Voce Examinations or absent for the assessments in case of courses evaluated only through continuous

assessments, this absence shall be considered as an appearance for the purpose of classification of degree.

17.4.8 If a student opts to drop PEC(s) as per the Clause 10.6, this appearance in the particular PEC shall be considered as an appearance for the purpose of classification of degree.

17.4.9 After successful completion of the courses as required for the award of degree, as per the curriculum, a consolidated grade sheet shall be printed and issued to all the students through respective Head of the Department.

17.4.10 The percentage of marks shall be printed in the consolidated grade sheet from the CGPA earned using,

% of Marks = CGPA x 10

17.4.11 Mandatory non-credit courses will not be taken into account for the calculation of CGPA.

17.5. **Photocopy and Revaluation**

A student can apply for photocopy of his/her semester examination answer paper in a theory course, as per the guidelines of the COE, on payment of a prescribed fee through proper application to the Controller of Examinations through the Head of the Institutions. The answer script is to be valued and justified by a faculty member, who has handled the subject and recommend for revaluation with the breakup of marks for each question. Based on the recommendation, the student can register for the revaluation through proper application to the Controller of Examinations. The Controller of Examinations will arrange for the revaluation and the results will be intimated to the student concerned through the Head of the Institutions. Revaluation is not permitted for practical courses and EEC courses. A student can apply for revaluation of answer scripts for not exceeding 5 subjects at a time.

17.6. **Review**

Students not satisfied with Revaluation can apply for Review of his/ her examination answer paper in a theory course, within the prescribed date on

payment of a prescribed fee through proper application to the Controller of Examinations through the Head of the Institution. Students applying for Revaluation only are eligible to apply for Review.

17.7. The grades O, A+, A, B+ and B or the grades in all special courses including Industry oriented courses additional Internship and Online courses shall figure in the Grade sheets. The courses for which the grades are RA and SA will not figure in the Grade sheet.

18. AWARD OF DEGREE AND CLASSIFICATION

18.1. ELIGIBILITY FOR THE AWARD OF DEGREE

A student shall be declared to be eligible for the award of the B.E./B.Tech. Degree provided the Student has

18.1.1. Successfully gained the required minimum number of total credits as specified in the curriculum corresponding to his / her programme within the stipulated time.

18.1.2. Successfully completed the course requirements, appeared for the End Semester Examinations and passed all the courses prescribed in all the eight semesters within a maximum period of seven years (six years in the case of lateral entry) from the commencement of the first (third in the case of lateral entry) semester to which the student was admitted.

18.1.3. Successfully passed any additional courses prescribed by the Course Equivalency Committee constituted by the Principal whenever re-admitted under any other regulations.

18.1.4. No disciplinary action is pending against him /her.

18.2. CLASSIFICATION OF THE DEGREE AWARDED

18.2.1 First Class with Distinction

A student who satisfies the following conditions shall be declared to have passed the examination in First class with Distinction:

- Should have passed the examination in all the courses of all the eight semesters (6 semesters in the case of Lateral Entry) in the student's First Appearance within five years (Four years in the case of Lateral Entry). Withdrawal from examination will not be considered as an appearance.
- Should have secured a CGPA of not less than 8.50.
- One year authorized break of study (if availed of) is included in the five years (four years in the case of lateral entry) for award of First class with Distinction.
- Should not have been prevented from writing end semester examination in any of the courses of the Curriculum making up the total credit requirement.
- A student who fulfills the criteria as prescribed in Honours Degree as per Clause 19 / Clause 20 shall be declared to have successfully completed the degree programme with **'Honours' / 'Minor'** and classified under **'First Class with Distinction'**.

18.2.2 **First Class**

A student who satisfies the following conditions shall be declared to have passed the examination in First class:

- Should have passed the examination in all the courses of all eight semesters (6 semesters in the case of Lateral Entry) within five years.
- One year authorized break of study (if availed of) or prevention from writing the End Semester examination due to lack of attendance (if applicable) is included in the duration of five years (four years in the case of lateral entry) for award of First class.
- Should have secured a CGPA of not less than 6.50.
- A student who fulfills the criteria as prescribed in Honours Degree as per Clause 19 / Clause 20 shall be declared to have successfully completed the degree programme with **'Honours' / 'Minor'** and classified under **'First Class'**.

18.2.3 **Second Class**

Students who are not covered in clauses 18.2.1 and 18.2.2 and who qualify for the award of the degree shall be declared to have passed the examination in Second Class.

18.2.4 A student who is absent in the End Semester Examination in a course / project work after having registered for the same shall be considered to have appeared in that examination (except approved withdrawal from end semester examinations) for the purpose of classification.

18.2.5 Student who earned an additional 18 credits as per Clause 19 but does not satisfy the conditions mentioned in 18.2.1 and 18.2.2 shall not be awarded B.E./B.Tech. (Honours). In such case if the student becomes eligible for First Class, while computing CGPA with the Professional Elective / Open Elective courses with higher grades the student shall be awarded B.E. / B.Tech. degree in First Class only.

18.3. A student who is absent in end semester examination in a course / project work after having registered for the same shall be considered to have appeared in that examination for the purpose of classification.

19. Honours Degree

19.1. Students always have a passion or fascination for a particular field or a line of thinking or study and Honours Degree option enables the student to nurture this interest from. Students shall obtain Honours degree by successful completion of additional courses over and above the courses prescribed in the respective curriculum from 'FIFTH' semester onwards. Additional courses shall be either a

'CAPSTONE DESIGN PROJECT' or 'SIX ELECTIVE COURSES'.

19.1.1. Students shall earn 18 credits and split up of credits shall be given in the respective curriculum.

19.1.2. Eligibility for any student to opt for the Honours degree is that, the student should earn a minimum GPA of 7.5 in 'THIRD' and 'FOURTH' semesters and also students should not have any history of arrears from first semester.

19.2. Capstone Design Project for 'HONOURS' Degree

19.2.1. Capstone Design project provides an opportunity to work in a team to design and develop a product, focusing to evolve solutions for real-time problems, starting from Problem definition, Specification formulation within given constraints, Review of design process and design ideation, Human factors, Market research, Product and patent research, fabrication considerations, Safety and risk assessment, Liability and ethics, Environmental and sustainability considerations, Proof-of-concept methods, Codes and Standards, etc. and finally to test prototype(s).

19.2.2. A group comprising of 4 to 6 students, spread over minimum of two departments and maximum of three departments, shall opt for Capstone Design Project in 'THREE PHASES' from V to VII semester, in addition to the courses as prescribed in the curriculum. It is mandatory that students from at least TWO departments shall be a part of the project group and the number of students from each department shall be at least TWO.

19.2.3. A Mentor shall be identified by the students from any one of the participating Department and Head of the Department of the Mentor's Department shall constitute a Project Review Committee (PRC) consisting of Mentor, one faculty – from the department of the students (other than the Mentor's Department) in the group and one expert from industry in the relevant field. Student shall also identify Co-Mentor(s) from the participating department(s).

19.2.4. During the Capstone Design Project, the student groups are encouraged to participate in events such as Hackathon, Industrial Expo to showcase and pitch their inventions.

19.2.5. Evaluation of Capstone Design Project is by continuous assessment (Three periodical reviews) in each semester by the PRC and Viva-voce examination at the end of each semester. Viva-Voce examination shall be conducted with panel of internal examiners (PRC) and ONE external examiner. Weightage for Continuous assessment is 60% and for project report evaluation along with Viva-voce examination is 40%.

19.2.6. Every student in the group shall secure a minimum 45% of marks in Project report evaluation & viva-voce examination and also obtain a 50% of total marks in the Continuous assessments + (Project report evaluation & viva-voce examination) for getting 'PASS' in the particular phase of Capstone Design Project. Upon successful completion of the particular phase, the students shall be allowed to proceed to the next phase. In case, if any student in the group is not able to secure 'PASS' as prescribed above, the student shall discontinue from the Capstone Design Project and shall not associate with the Capstone Design Project furthermore.

19.2.7. After earning 18 credits through the successful completion of the Capstone Design Project (CDP), 'HONOURS' shall be conferred along with 'Certificate of Appreciation' to the students in this group, who secure 8.50 or higher CGPA, without any history of arrears. If the student in the group, after successful completion of Capstone Design Project secures less than 8.50 CGPA, 'HONOURS' degree shall not be conferred. However, the result of Capstone Design Project shall feature in the Grade sheet and a 'Certificate of Appreciation' shall be issued to those students.

19.3. **Six Elective Courses for 'HONOURS' Degree**

19.3.1. Students shall opt for 'Honours' Degree by undergoing 'SIX' additional elective courses from the Honours vertical or courses from existing Professional Elective verticals from V to VII semester as listed in the respective curriculum. The student has to earn 18 additional credits through the course works, which are related to the major discipline of study. Evaluation and 'PASS' requirement of these courses is as per Clauses 12.1, 12.2 and 16 for Theory & Laboratory Integrated Theory courses respectively.

19.3.2. On successful completion of six elective courses, 'HONOURS' shall be conferred to the students, who secure a CGPA of 8.50 or higher, without any history of arrears. If the student after successful completion of 'SIX' courses secures less than 8.50 CGPA, 'HONOURS' degree shall not be conferred.

19.3.3. If the student decides to opt out of Honours Degree, after successful completion of certain number of course work(s), these course(s) shall be treated as equivalent to PEC(s). In such case, the student shall be permitted to drop the equal number of PEC(s).

19.3.4. If the student successfully completes more number of PECs over and above the SIX mandatory courses, the courses with higher grades shall be considered for the calculation of CGPA. Remaining courses shall be only printed on the grade sheet and grade obtained in these courses shall not be considered for computation of CGPA.

20. Minor Degree

20.1. Eligibility for any student to opt for the Minor degree is that the student should earn a minimum GPA of 7.50 in III and IV semesters.

20.2. A student can register for a 'Minor' degree, from the V semester by registering basket of courses as prescribed in the curriculum of the UG Programme offered by the department other than the parent department of the student. Evaluation and 'PASS' requirement of these courses is as per Clauses 12.1, 12.2 and 16 for Theory & Laboratory Integrated Theory courses respectively.

20.3. In order to get a Minor Degree, a student has to successfully complete the basket of courses as prescribed in the curriculum of the Department offering the Minor Degree and earn 18 extra credits. Grades obtained in the basket of courses shall also be considered for computation of CGPA, along with the courses in the curriculum for the programme (mainstream).

20.4. If the student decides to opt out of Minor Degree, after successful completion of certain number of course(s), these course(s) can be treated as equivalent to OEC(s). In such case, the student shall be permitted to drop the equal number of OEC(s).

20.5. If the student successfully completes more number of OECs over and above the courses prescribed in the curriculum, the courses with higher grades shall be considered for the computation of CGPA. Remaining courses shall be only printed on the grade sheet and grade obtained in

these courses shall not be considered for computation of CGPA.

20.6. Minor Degree shall be reflected in the degree certificate as “B. E., / B. Tech., in XYZ Engineering with Minor in ABC”, where ‘XYZ’ is the Major Discipline of study and ‘ABC’ is the Discipline in which Minor Degree is offered.

Note: The students shall be permitted to opt for either ‘Honours’ or ‘Minor’ Degree, during the programme.

21. PROVISION FOR WITHDRAWAL FROM EXAMINATIONS

21.1. A student may, for valid reasons, be granted permission to withdraw from any or all current semester examinations. Application for withdrawal is permitted ONLY ONCE in the entire duration of the programme.

21.2. The withdrawal application is valid only if the student is otherwise eligible to write the examination and is submitted to the Principal after duly recommended by the Head of the Department, prior to the last examination of that semester.

21.3. The withdrawal shall not be considered as an appearance for deciding the classification for First Class with Distinction.

21.4. A final semester student who has withdrawn may be permitted to appear for immediate examinations to be conducted within a reasonable time.

21.5. The final semester student who has withdrawn from appearing for project viva-voce for genuine reasons shall be permitted to appear for immediate viva-voce examination within a reasonable time with proper application approved by the Principal submitted to the Controller of Examinations.

22. AUTHORIZED BREAK OF STUDY FROM A PROGRAMME

22.1. A student is permitted to avail an Authorized Break of Study under valid reasons (such as accident, hospitalization, prolonged ill health, etc.,) and to rejoin the programme in a later semester. The student shall apply in advance to the Principal, through the Head of the Department, stating the reasons, in the beginning of the semester.

22.2. A student is permitted to avail the authorized break of study, for a maximum period ofone year, ONLY ONCE during the entire period of study.

22.3. The student permitted to rejoin the programme after break shall be governed by the rulesand regulations in force at the time of rejoining.

22.4. The total period of completion of the programme, reckoned from the commencement of the semester to which the student was admitted, shall not exceed the maximum period specified in clause 6 including of the period of break of study in order to qualify for the award of the degree.

23. DISCIPLINE

23.1. Every student is required to observe disciplined and decorous behaviour both inside and outside the college and not to indulge in any activity which will tend to bring down the prestige of the University / College. The Head of the Institution shall constitute a disciplinary committee consisting of the Head of the Institution, Two Heads of Department of which one should be from the faculty of the student, to enquire into acts of indiscipline and notify the University about the disciplinary action recommended for approval. In case of any serious disciplinary action which leads to suspension or dismissal, then a committee shall be constituted including one representative from Anna University, Chennai. In this regard, the member will be nominated by the University on getting information from the Head of the Institution.

23.2. If a student indulges in malpractice in any of the University / internal examination he / she shall be liable for punitive action as prescribed by the University from time to time.

24. SPECIAL CASES

In the event of any clarification in the interpretation of the above rules and regulations, they shall be referred to the Standing Committee. The Standing Committee will offer suitable interpretations/ clarifications/amendments required for the special case on such references and get them ratified in the next meeting of the Academic Council. The decision of the Academic Council will be final.

25. REVISION OF REGULATIONS, CURRICULUM AND SYLLABI

Notwithstanding anything contained in this Regulation, the Academic Council of the College, reserves the right to modify/amend, the provisions of these Regulations, Curriculum, Syllabus, procedures, requirements, and rules pertaining to its undergraduate programmes.

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ANNEXURE – 1

LIST OF QUALIFYING EXAMINATIONS FOR NRI/FOREIGN NATIONALS/ PIO /OCI / CIWGC

Candidates must have passed the qualifying examination i.e (10+2 pattern) of Indian System or its equivalent (completed 12 years of school education). Candidates must have studied in English medium. The candidate should have passed in all subjects and scored a minimum average of 45 percentage in Mathematics, Physics and Chemistry put together in qualifying examinations for admissions in FN/PIO/OCI/CIWGC and for admissions in NRI under General Category (OC). For admissions in NRI under reserved categories including BC/BCM/MBC/DNC/SC/SCA/ST, the minimum average must be 40 percentage in Mathematics, Physics and Chemistry put together in qualifying examinations.

List of Qualifying Examinations equivalent to Indian System

- GCE Examination of U.K. with pass grades in five subjects at the Ordinary Level (O level) and two subjects at the Advanced Level (A Level).
- Higher School Certificate Examination, University of Cambridge Local Examinations Syndicate, U.K.
- International Baccalaureate Examination of Geneva. High School Graduation from accredited Institutions of USA.
- 12 Years High School Diploma from China
- 12 Years High School Graduation Diploma of International School, Bangkok / Grade 12 Examination of Ruam Rudee International School, Bangkok / 12 Years High School Diploma of Adventist English School, Bangkok Mathayom VIII, Final Examination of Upper Secondary Education Course; Final Examination of Secondary Education Course; Mathayom Suksa, 5 Pre-University Final Examination (12 Year Course) in Ministry of Education, Thailand.
- 12 Years High School Graduation Diploma from Myanmar

- 12 Years Bhutan Higher Secondary Education Certificate
- 12 Years High School Graduation Diploma from Syria
- 12 Years of Secondary Stage qualifications of National Examination Board / Council of Ministries of Education from Kenya, Nigeria, Sudan, Tanzania and Uganda
- 12 Years General Secondary School Certificate of Ministry of Education, Sultanate of Oman and Yemen
- SMA/SLTA/Senior High School/Secondary School Advanced stage exams from Indonesia
- Intermediate Science Course, Kathmandu University, Kathmandu, Nepal
- Intermediate / Senior Secondary / Higher Secondary / 12 Years High School Diploma /
- Certificate Level Examination of the approved Board / Council of Secondary Education from Bangladesh, Sri Lanka, Iran, Iraq, Bahrain, Kuwait, Jordan, Saudi Arabia, UAE and other Gulf Countries.
- Sijil Tinggi Persekolahan Malaysia (STPM) of Ministry of Education, Malaysia / Matriculation Science Course of National University of Malaysia / 12 Years Matriculation Science Certificate of University of Malaya & University of Science, Malaysia.